Newton's Laws

ClipSAT Printable Practice Suite (clipsat.org) • Topic 13.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Particles of mass 11 kg and 16 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 127.76. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 1.81$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 11(9.8 + 1.81) = 127.76$ N

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Newton's Laws of Motion · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Express $\frac{3 + \sqrt{5}}{2 - \sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are rational numbers to be found.

Multiply numerator and denominator by the conjugate of the denominator, $2 + \sqrt{5}$:
$\frac{3 + \sqrt{5}}{2 - \sqrt{5}} \times \frac{2 + \sqrt{5}}{2 + \sqrt{5}} = \frac{(3 + \sqrt{5})(2 + \sqrt{5})}{2^2 - (\sqrt{5})^2}$ Expanding the numerator: $3(2) + 3\sqrt{5} + 2\sqrt{5} + (\sqrt{5})^2 = 6 + 5\sqrt{5} + 5 = 11 + 5\sqrt{5}$.
The denominator is $4 - 5 = -1$. Therefore: $\frac{11 + 5\sqrt{5}}{-1} = -11 - 5\sqrt{5}$ Thus $p = -11$ and $q = -5$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A circle has radius 8 cm. A sector of this circle has a total perimeter of 28 cm. Find the angle of the sector in radians, and calculate the exact area of the sector.

The perimeter of a sector is given by $P = s + 2r = r\theta + 2r$. Substitute $P = 28$ and $r = 8$:

$28 = 8\theta + 2(8) \Rightarrow 28 = 8\theta + 16 \Rightarrow 8\theta = 12 \Rightarrow \theta = \frac{12}{8} = 1.5$ radians The area of the sector is:

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(8^2)(1.5) = \frac{1}{2}(64)(1.5) = 32 \times 1.5 = 48$ cm^2

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

Find the critical values where $2x^2 - 7x + 3 = 0$. Factor the quadratic: $(2x - 1)(x - 3) = 0$, so the roots are $x = \frac{1}{2}$ and $x = 3$. Since the coefficient of x^2 is positive ($a = 2 > 0$), the parabola opens upward. The quadratic is less than or equal to zero between and including the roots. Therefore, the solution set is $\frac{1}{2} \leq x \leq 3$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x + 3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.

(a) $y = f(x + 3) + 2$ translates 3 units left and 2 units up: $(4 - 3, -6 + 2) = (1, -4)$. (b) $y = 3f(2x)$ stretches horizontally by $\frac{1}{2}$ and vertically by 3: $\left(4 \times \frac{1}{2}, -6 \times 3\right) = (2, -18)$. (c) $y = -f(-x)$ reflects in both axes (point reflection through the origin): $(-4, -(-6)) = (-4, 6)$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student computes the volume of a cylinder using $r = 3.2$ cm and $h = 7.85$ cm. The raw calculator display shows $252.518...$. How should the final answer be reported according to Cambridge AS Level guidelines?

Unless otherwise specified, non-exact numerical answers must be given to 3 significant figures. Rounding $252.518...$ to 3 s.f. gives $253\ \text{cm}^3$. Giving 1 or 2 s.f., or excessive decimal places without justification, incurs an accuracy penalty (A1).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A discrete random variable Y has $E(Y) = 5$ and $\text{Var}(Y) = 4$. Calculate $E(Y^2)$ and $\text{Var}(10 - 3Y)$.

Using $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$, we get $E(Y^2) = \text{Var}(Y) + [E(Y)]^2 = 4 + 5^2 = 4 + 25 = 29$. For the linear transformation: $\text{Var}(10 - 3Y) = (-3)^2\text{Var}(Y) = 9 \times 4 = 36$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A continuous dataset is grouped into class intervals. The interval $20 \le x < 35$ has a frequency of 45. Find the frequency density for this class.

Class width $w = 35 - 20 = 15$. Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{45}{15} = 3.0$. In a histogram, the height of the bar drawn on the interval $[20, 35)$ is 3.0.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the number of ways to arrange the 7 letters in the word SUCCESS.

The word SUCCESS has 7 letters with 3 S's, 2 C's, 1 U, and 1 E. The number of distinct permutations is $\frac{7!}{3!\,2!} = \frac{5040}{6 \times 2} = \frac{5040}{12} = 420$.

© ClipSAT — clipsat.org | All Rights Reserved

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Indices, Surds & Quadratics · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Differentiation & Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Simultaneous Equations & Inequalities · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Trigonometry · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.01 Indices, Surds & Quadratics**

[About AS Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 10.01 Series & Binomial Theorem**

[Coordinate Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

© ClipSAT — clipsat.org | All Rights Reserved

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Kinematics

[Representation of Data · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 12.01 Forces & Equilibrium

[Discrete Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 14.01 Energy, Work & Power

[Energy, Work & Power · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 15.01 Representation of Data

[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A constant resultant force of 36 N acts on a mass of 4 kg. What is the acceleration produced?

a. $9\ \text{m/s}^2$ ✓

b. $144\ \text{m/s}^2$

c. $4\ \text{m/s}^2$

d. $6\ \text{m/s}^2$

$a = F/m = 36/4 = 9\ \text{m/s}^2$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A car of mass 1200 kg is traveling at 25 m/s when the brakes are applied, bringing it to rest in 50 m. Find the magnitude of the braking force.

a. 7500 N ✓

b. 6000 N

c. 9000 N

d. 15000 N

$v^2 = u^2 + 2as \Rightarrow 0 = 25^2 + 2a(50) \Rightarrow 100a = -625 \Rightarrow a = -6.25$ m/s². Force magnitude $F = ma = 1200(6.25) = 7500$ N.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An elevator of mass 800 kg accelerates upwards at 1.5 m/s². Taking $g = 10$ m/s², find the tension in the cable.

a. 9200 N ✓

b. 8000 N

c. 6800 N

d. 1200 N

Equation of motion: $T - mg = ma \Rightarrow T = m(g + a) = 800(10 + 1.5) = 800(11.5) = 9200$ N.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two masses of 3 kg and 2 kg are connected by a light inextensible string passing over a smooth fixed pulley (Atwood machine). Taking $g = 10$ m/s², find the acceleration of the system.

a. 2 m/s² ✓

b. 1 m/s²

c. 4 m/s²

d. 5 m/s²

Equations: $3g - T = 3a$ and $T - 2g = 2a$. Adding: $(3 - 2)g = (3 + 2)a \Rightarrow 10 = 5a \Rightarrow a = 2$ m/s².

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the system in the previous question, find the tension T in the string.

a. 24 N ✓

b. 25 N

c. 20 N

d. 30 N

$T = 2(g + a) = 2(10 + 2) = 2(12) = 24$ N.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the total force exerted by the string on the pulley in the previous system?

a. 48 N **downwards** ✓

b. 24 N downwards

c. 50 N downwards

d. 40 N downwards

The string pulls downward on both sides of the pulley with tension $T = 24$ N. Total force on pulley $= 2T = 48$ N downwards.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A block of mass 5 kg is pulled up a smooth slope inclined at 30° to the horizontal by a force of 40 N acting parallel to the slope. Taking $g = 10$ m/s^2, find the acceleration of the block.

a. 3 m/s^2 ✓

b. 5 m/s^2

c. 2 m/s^2

d. 4 m/s^2

$F_{\text{net}} = P - mg\sin(30^\circ) = 40 - 5(10)(0.5) = 40 - 25 = 15$ N. Then $a = 15/5 = 3$ m/s^2.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the slope in the previous question is rough with $\mu = 0.2$, what is the new acceleration of the block?

a. 1.27 m/s^2 ✓

b. 3.00 m/s^2

c. 2.15 m/s^2

d. 0.85 m/s^2

$R = mg\cos(30^\circ) = 50(\sqrt{3}/2) \approx 43.30$ N. Friction $F_f = \mu R = 0.2(43.30) \approx 8.66$ N. Net force $= 40 - 25 - 8.66 = 6.34$ N. Acceleration $a = 6.34/5 \approx 1.27$ m/s^2.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A sled of mass 20 kg slides down an incline of 25° at constant speed. What is the coefficient of kinetic friction μ?

a. $\mu = \tan(25^\circ) \approx 0.466$ ✓

b. 0.423

c. 0.906

d. 0.500

At constant speed, acceleration is zero, so $mg\sin\alpha = \mu mg\cos\alpha \Rightarrow \mu = \tan\alpha = \tan(25^\circ) \approx 0.466$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A block A of mass 4 kg rests on a smooth horizontal table and is connected by a light string over a smooth pulley at the edge of the table to a hanging block B of mass 1 kg. Taking $g = 10\ \text{m/s}^2$, find the acceleration.

a. $2\ \text{m/s}^2$ ✓

b. $2.5\ \text{m/s}^2$

c. $1.5\ \text{m/s}^2$

d. $10\ \text{m/s}^2$

Net accelerating force is the weight of B: $m_B g = 1(10) = 10$ N. Total mass $= 4 + 1 = 5$ kg. Acceleration $a = 10/5 = 2\ \text{m/s}^2$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the table in the previous system has coefficient of friction $\mu = 0.15$, find the acceleration of the system.

a. $0.8\ \text{m/s}^2$ ✓

b. $1.2\ \text{m/s}^2$

c. $1.5\ \text{m/s}^2$

d. $0.5\ \text{m/s}^2$

Friction on A: $F_f = \mu m_A g = 0.15(4)(10) = 6$ N. Net accelerating force $= 10 - 6 = 4$ N. Total mass $= 5$ kg. Acceleration $a = 4/5 = 0.8\ \text{m/s}^2$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A car of mass 1000 kg tows a trailer of mass 500 kg with acceleration 1.2 m/s^2. Resistance to motion is 300 N on the car and 200 N on the trailer. Find the tension in the towbar.

a. 800 N ✓

b. 600 N

c. 1100 N

d. 1500 N

Consider the trailer alone: $T - R_{\text{trailer}} = m_{\text{trailer}}a \Rightarrow T - 200 = 500(1.2) \Rightarrow T - 200 = 600 \Rightarrow T = 800$ N.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A resultant force acts on a body of mass 8 kg, giving it an acceleration of 8 m/s^2. Find the magnitude of the resultant force.

a. 62

b. **64** ✓

c. 128

d. 66

Applying Newton's Laws of Motion principles, $F = ma = 8 \times 8 = 64$ N Therefore, the verified correct option is 64.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A body of mass 4 kg is acted on by a driving force of 13 N and a resisting force of 1 N. Find its acceleration.

a. 2

b. 13/4

c. **3** ✓

d. 4

Net force $= 13 - 1 = 12$ N. $a = \frac{F}{m} = \frac{12}{4} = 3$ m/s^2

© ClipSAT — clipsat.org | All Rights Reserved

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Balls are drawn one at a time, without replacement, from a bag of 20 balls; X is the number of red balls drawn. Is X binomially distributed?

a. Yes — but only if $n > 30$

b. Yes — fixed, independent trials with constant success probability

c. **No — trials are not independent / probability is not constant ✓**

d. No — the number of trials is not fixed

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)?

a. **120 ✓**

b. 103

c. 122

d. 100

Applying Permutations & Combinations principles, $(n-1)! = (6-1)! = 120$ Therefore, the verified correct option is 120.

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

An engine exerts a driving force of 414 N on a vehicle moving at a constant 19 m/s. Find the power developed.

a. **7866 ✓**

b. 7885

c. 15732

d. 7847

Applying Energy, Work & Power principles, $P = Fv = 414 \times 19 = 7866$ W Therefore, the verified correct option is 7866.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 12.01 Forces & Equilibrium

Find the number of distinct arrangements of the letters of the word PUPPY.

a. 103

b. 27

c. 64

d. **60 ✓**

Applying Permutations & Combinations principles, $\frac{5!}{3!} = 60$ Therefore, the verified correct option is 60.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 14.01 Energy, Work & Power

A random variable $X \sim B(100, 0.6)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 58.0

b. 65.0

c. **60 ✓**

d. 62.0

Applying Normal Distribution principles, Mean $= np = 100 \times 0.6 = 60$ Therefore, the verified correct option is 60.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 15.01 Representation of Data

From a cumulative frequency curve with $n = 80$, the median is read at cumulative frequency:

a. 20

b. 40 ✓

c. 60

d. 80

Applying Representation of Data principles, Median at $n/2 = 40$ Therefore, the verified correct option is 40.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 16.01 Permutations & Combinations

Find the x-coordinates of the points of intersection of $y = x^2 - 9x + 22$ and $y = 2$.

a. **x=4 or x=5 ✓**

b. x=-4 or x=-5

c. x=5 or x=5

d. x=4 or x=4

Substituting the line into the curve gives $x^2 - 9x + 20 = 0$, which factorises as $(x - 5)(x - 4) = 0$, so $x = 4$ or $x = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 17.01 Discrete Random Variables

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -6)$. State the coordinates of the new vertex.

a. (-2,6)

b. **(-2,-6)** ✓

c. (-6,-2)

d. (2,-6)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-2,\ -6)$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 18.01 Binomial Distribution

The minimum point of $y = x^2$ is translated by the vector $(-3; 6)$ to give curve C. State the coordinates of the minimum point of C.

a. (3,6)

b. (6,-3)

c. **(-3,6)** ✓

d. (-3,-6)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3, 6)$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 19.01 Normal Distribution

The curve $y = x^2$ is translated by the vector $(7; 4)$. State the coordinates of the new vertex.

a. (-7,4)

b. **(7,4)** ✓

c. (4,7)

d. (7,-4)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(7, 4)$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the variance of the data set: $4, 8, 14, 9, 2, 6$.

a. 1/2

b. 1/4

c. 2/3

d. **533/36** ✓

Mean $= \frac{43}{6} = 43/6$. Variance $= \frac{\sum x^2}{n} - \bar{x}^2 = \frac{397}{6} - (43/6)^2 = 533/36$

© ClipSAT — clipsat.org | All Rights Reserved